

Lab 05

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```
#Lendo os pacotes
library(tidyverse)
library(ggplot2)
library(leaflet)
library(ggcal)
```

```
#Criando a Funcao com os argumentos tail_number, arquivo
```

```
analisa_aeronave <- function(tail_number, arquivo) {
  #Lendo os dados em chunks e ja aplicando o filtro de acordo com o tail_number
  dados <- read_csv_chunked(
    file = arquivo,
    callback = DataframeCallback$new(\(chunk, pos) {
      chunk |> filter(TAIL_NUMBER == tail_number)
    }),
    chunk_size = 1e5,
    show_col_types = FALSE
  )

  #Obtendo os dados no numero da cauda
  tail_data <- dados |>
    #Ordenando por data e hora
    arrange(YEAR, MONTH, DAY, DEPARTURE_TIME) |>
    #Juntando DAY, MONTH e YEAR em uma coluna unida chamada DATA no formato DD/MM/AAAA
    mutate(DATA = sprintf("%02d/%02d/%04d", DAY, MONTH, YEAR), .before = YEAR)

  #Retornando o valor esperado da funcao
  return(tail_data)
}
```

```
#Teste com o voo do exemplo N431WN
analisa_aeronave("N431WN", "C:/Users/joaoz/Desktop/Lab05/flights/flights.csv")
```

```
# A tibble: 1,938 x 32
```

	DATA	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	01/01/2015	2015	1	1	4	WN	1762	N431WN
2	01/01/2015	2015	1	1	4	WN	3564	N431WN
3	01/01/2015	2015	1	1	4	WN	3564	N431WN
4	01/01/2015	2015	1	1	4	WN	3564	N431WN
5	02/01/2015	2015	1	2	5	WN	1793	N431WN
6	02/01/2015	2015	1	2	5	WN	1793	N431WN
7	02/01/2015	2015	1	2	5	WN	1793	N431WN
8	02/01/2015	2015	1	2	5	WN	1793	N431WN
9	02/01/2015	2015	1	2	5	WN	1793	N431WN
10	02/01/2015	2015	1	2	5	WN	3144	N431WN

```
# i 1,928 more rows
```

```
# i 24 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
# SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
# TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
# AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
# SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
# DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-09-17 02:26:48 -03"
```